



SpectroCVT-Net: A convolutional vision transformer architecture and channel attention for classifying Alzheimer's disease using spectrograms

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ABSTRACT

Dementia arises from various brain-affecting diseases and injuries, with Alzheimer's disease being the most prevalent, impacting around 55 million people globally. Current clinical diagnosis often relies on biomarkers indicative of Alzheimer's distinctive features. Electroencephalography (EEG) serves as a cost-effective, user-friendly, and safe biomarker for early Alzheimer's detection. This study utilizes EEG signals processed with Short-Time Fourier Transform (STFT) to generate spectrograms, facilitating visualization of EEG signal properties. Leveraging the Brainlat database, we propose SpectroCVT-Net, a novel convolutional vision transformer architecture incorporating channel attention mechanisms. SpectroCVT-Net integrates convolutional and attention mechanisms to capture local and global dependencies within spectrograms. Comprising feature extraction and classification stages, the model enhances Alzheimer's disease classification accuracy compared to transfer learning methods, achieving $92.59 \pm 2.3\%$ accuracy across Alzheimer's, healthy controls, and behavioral variant frontotemporal dementia (bvFTD). This article introduces a new architecture and evaluates its efficacy with unconventional data for Alzheimer's diagnosis, contributing: SpectroCVT-Net, tailored for EEG spectrogram classification without reliance on transfer learning; a convolutional vision transformer (CVT) module in the classification stage, integrating local feature extraction with attention heads for global context analysis; Grad-CAM analysis for network decision insight, identifying critical layers, frequencies, and electrodes influencing classification; and enhanced interpretability through spectrograms, illuminating key brain wave contributions to Alzheimer's, frontotemporal dementia, and healthy control classifications, potentially aiding clinical diagnosis and management.

1. Introduction

Dementia results from a variety of diseases and injuries that affect the brain. Alzheimer's disease is the most common form of dementia, affecting around 55 million people worldwide, with more than 60% of them living in low- and middle-income countries, generating millions of new cases annually. Dementia is a syndrome that can be caused by various diseases that, over time, destroy nerve cells and damage the brain, typically leading to a deterioration in cognitive function beyond what might be expected from the usual consequences of biological aging. This has physical, psychological, social, and economic impacts, not only for people living with dementia but also for their caregivers,

families, and society at large [1]. Clinical diagnosis today commonly involves biomarkers, which indicate that the hallmark characteristics of Alzheimer's disease (AD) include the presence of neurofibrillary tangles, senile plaques, and significant neuronal cell loss in the hippocampus [2]. In the rise of deep learning (DL) mechanisms in health fields such as tumor detection, strokes, COVID-19, among others [3–6], it is important that these mechanisms not only remain in the literature but also be tested in clinical environments to contribute to patient well-being, diagnostic accuracy, and the reduction of time and costs associated with these pathologies. This opens the opportunity for the use of new technologies in clinical environments that generate a real impact, including AD.

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Electroencephalography (EEG) is currently used as a biomarker in the early diagnosis and detection of AD and other types of dementia. Additionally, this technique has advantages over other brain data collection techniques, such as MRI and fMRI, since EEGs measure brain activity directly. EEGs are comparatively less expensive than MRI and fMRI, require less training since they are easier to use, and are safer as there is no radiation involved. As mentioned earlier, low- and middle-income countries are the most affected by this pathology. The difficult access to advanced equipment gives EEG another advantage, as EEG machines are portable or semi-portable [7]. Therefore, EEG plays an essential role in meeting these needs and is a valuable tool for detecting AD. Additionally, visual and statistical techniques can be applied to EEG data for diagnosing dementia [8,9].

In this paper, EEG signals were used with a spectrum estimation technique called Short-Time Fourier Transform (STFT). This technique uses spectrograms to convert the temporal information of the EEG into time and frequency information, allowing for the visualization of more properties and characteristics that the temporal form of the EEG does not reveal. Recently, this spectrogram technique has been used in classification tasks [10–15], demonstrating great interpretability in determining significant frequencies and the cerebral regions from which the model extracts the most information. This helps to identify which brain waves [16] and lobes contribute more information in classification.

In this study, the Brainlat database was used, which is a multimodal set of neuroimages including MRI, fMRI, DWI, and EEG, from 780 patients with neurodegenerative diseases such as AD, behavioral variant frontotemporal dementia (bvFTD), multiple sclerosis (MS), Parkinson's disease (PD), and healthy controls (HC). This article specifically selected AD, bvFTD, and HC classes for this investigation and worked with electroencephalographic (EEG) signals. In total, 35 AD patients, 19 bvFTD patients, and 32 HC patients containing EEG signals were included. A brief description of each stage is presented below:

- **AD:** At this stage, the patient requires constant accompaniment, since he/she loses spatio-temporal awareness, presents behavioral disorders and aggressiveness, and even forgets close relatives.
- **bvFTD:** It is a disease that manifests through subtle and gradual changes in personality and interpersonal behavior, indicating a progressive disintegration of neural circuits involved in social cognition, emotional regulation, motivation, and decision-making.
- **HC:** Refers to patients without dementia.

Although the currently existing Convolutional Neural Network (CNN) methods for classifying EEG signals show promising results [11,12,17–20], the present work is looking to solve the problem of interpretability.

In this research, pretrained CNNs with ImageNet [21], such as VGG16, VGG19, InceptionResNetV2, InceptionV3, DenseNet201, EfficientNetB7 and ConvNeXtXLarge, based on convolutional neural networks, were used to compare with the proposed architecture SpectroCVT-Net. These CNN-based methods only consider the local features of images, and the spectrogram's global relations are not taken into account. The proposed neural network comprises a feature extraction stage and a classification stage. This architecture combines local (convolutional layers) and global (Vision Transformer) spectrogram ratios in the same architecture that integrates convolutional layers for local feature extraction with Vision Transformer (ViT) layers for global feature modeling.

The network architecture consists of two stages: extraction and classification. In the extraction stage, spectrograms of size $224 \times 224 \times 128$ are processed. This stage uses a convolutional block that includes multiple Conv2D layers followed by batch normalization and LeakyReLU activation functions, allowing for reduced spatial dimensions while increasing the depth of the feature maps. In addition, a skip connection is employed to prevent gradient fading regardless of neural network

depth. SE-Blocks are also incorporated, which improve the feature maps by modeling interdependencies between channels, resulting in an improved feature representation. The classification stage includes a Convolutional ViT consisting of multi-head auto attenuation layers, layer normalization, and dense layers. This allows the model to capture complex patterns by attending to different parts of the spectrogram globally.

The proposed neural network architecture achieved an accuracy of $92.59 \pm 2.3\%$ for the classification of AD, HC, and bvFTD; $98.51 \pm 2.9\%$ for AD vs HC; $100 \pm 0\%$ for bvFTD vs HC and $85 \pm 4.4\%$ for AD vs bvFTD, demonstrating its effectiveness in EEG spectrogram classification. These results underscore the effectiveness of our neural network architecture in classifying EEG spectrograms and its potential utility in the clinical diagnosis of neurological conditions such as Alzheimer's Disease and frontotemporal dementia.

The main contributions of this work are summarized as follows:

- A network for extracting local and global features from spectrograms called SpectroCVT-Net is proposed. This network, specifically designed for EEG spectrogram classification, is not based on transfer learning. It consists of two stages: feature extraction and classification, tailored to capture the intricate details of EEG data.
- The classification stage employs a convolutional vision transformer (CVT) module. This innovative module integrates convolutional layers for local feature extraction and multiple attention heads to capture global relationships within spectrograms. By attending to different parts of the spectrogram image, the CVT module effectively analyzes the complete global context. The main contribution of this architecture is its ability to capture dependencies and long-term relationships between different positions or locations within the image using the attention mechanism.
- Grad-CAM analysis was used to better understand the network decision-making process. This technique allowed us to identify which layers and frequencies contribute most to the classification of each class and also helps to determine which electrode is most influential in each disease.
- Spectrograms provide greater interpretability by allowing us to identify which brain waves contribute most to the classification of each class. This allows a deeper understanding of the neural underpinnings associated with Alzheimer's disease, frontotemporal dementia, and healthy controls, potentially aiding in the clinical diagnosis and follow-up of these conditions.

The remainder of the document is organized as follows: Section 2 has a concise overview of previous CNN architectures. Section 3 provides details of the processing and databases used. Section 4 presents a new architecture called SpectroCVT-Net, which covers all stages: feature extraction and classification. It also specifies the hyperparameters of SpectroCVT-Net. Section 5 describes the architectures used to compare with SpectroCVT-Net. Section 5.3 describes the metrics used in this study. Section 5.4 describes the software and hardware required. Section 6 includes tables and figures showing and discussing the performance of SpectroCVT-Net. Finally, Section 8 summarizes the study's conclusions.

2. Related work

Dileep et al. in their study about the AD detection process using EEG signals, involves implementing a deep learning strategy with the Optimized Transformer-based Attention Long Short Term Memory (OTA-LSTM) model. This model utilizes EEG signals decomposed with the Lifting Wavelet Transform (LWT), extracting temporal features with a Recurrent Neural Network (RNN) and spatial features with a Multi-scale dilated CNN. The Enhanced Wild Geese Lemurs Optimizer (EWGLO) algorithm is then applied to find optimal weight values for the weighted stacked features used in the OTA-LSTM model for AD

detection. The model achieved an accuracy of 96% and a Matthews Correlation Coefficient (MCC) of 98% [22].

Ozlem et al. use various signal decomposition methods, including Empirical Mode Decomposition (EMD), Ensemble EMD (EEMD), and Discrete Wavelet Transform (DWT), to distinguish EEG segments of control subjects and Alzheimer's dementia patients. Time-domain features were calculated using selected Intrinsic Mode Functions (IMFs) and detail and approximation coefficients of DWT. Different classification techniques such as Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Random Forest (RF) were employed to classify the EEG segments. The highest classification accuracy obtained using the DWT method for distinguishing control subjects and AD patients was 97.40% from the temporal/left brain cluster using the RF classifier [10].

Ozlem et al.'s make a investigation for classifying Alzheimer's dementia EEG involves obtaining joint time-frequency representations of AD and control subjects (CS) EEG segments, followed by deep feature extraction and classification using various classifiers such as DT, SVM, k-NN, and RF. The deep feature extraction is performed using Resnet-18 CNN architecture. The highest performance in distinguishing EEG segments of control subjects and AD patients was achieved with 89.84% accuracy and 91.56% F1-Score from the Temporal/Left brain cluster using the Short-Time Fourier Transform (STFT) method and k-NN classifier [11].

Sy sen et al. utilized a 1D CNN model trained on Proper Rotation Components (PRCs) extracted from EEG segments using the Intrinsic Time Scale Decomposition (ITD) method. The results showed that the ITD-based approach outperformed using plain EEG signals for classification, with an accuracy of 97.00% for EEG-PRCs classification compared to 79.50% for plain EEG segment classification. The sensitivity and specificity were also notably higher for the EEG-PRCs classification approach. It was highlighted that the proposed method demonstrated high classification accuracy in detecting EEG segments of AD patients using the ITD method with a 1D CNN approach. The results indicated the effectiveness of leveraging ITD and deep learning for Alzheimer's dementia classification based on EEG signals [12].

Jie-Min et al. developed an MFCC-based feature for epilepsy detection and the reliability of this feature through experiments. The method using MFCC-based features and XGBoost achieved a high accuracy of 99.5% in epilepsy detection; the XGBoost classifier using MFCC-based features obtained high accuracy in detecting epilepsy. The method achieved very high accuracy levels close to the state-of-the-art methods on both the three-class task and the two-class task; the technique of MFCC-based features and XGBoost can achieve very high accuracy in epilepsy detection, reaching levels comparable to state-of-the-art methods. The similarities between speech and EEG signal processing methods indicate the potential for good performance in processing EEG signals [13].

Mohamed et al. to diagnose AD using EEG data and deep neural networks classification, developed a new method for improving the classification. The technique involved creating a low-cost EEG device and utilizing a deep neural network to classify patients into three classes: MCI, AD, and healthy. The EEG data was collected during a 3-level N-Back working memory test, known as event-related potential (ERP), and then processed for classification. The proposed method in the document included two main parts: EEG electrode placement and data collection, focusing on brain waves in theta, alpha, and beta frequencies and transforming the data for classification using a CNN. The results showed a classification accuracy of 90.36% for healthy vs. MCI patients and 92.56% for healthy vs. AD patients, indicating consistent results compared to related work. It was highlighted that early diagnosis of AD is crucial to slow down its progression, reduce healthcare costs, and provide patients with extra time. The proposed system for early detection of AD was based on detecting specific biomarkers in EEG signals, such as ERP analysis and frequency sub-bands of brain waves, to train a CNN for classification [17].

Ghina et al. to classify AD, used a methodology that integrates EEG data with ML algorithms. The process includes EEG data collection, signal preprocessing, feature extraction, feature selection, and classification. The conclusion highlights the importance of collecting high-quality data, appropriate signal cleaning, selecting relevant features, and accurately classifying the disease. This approach can lead to the development of non-invasive and effective technologies for early diagnosis and intervention of AD. The results obtained from this methodology show promising potential for combining EEG with ML to classify AD accurately. By following a systematic process of EEG data analysis and classification, researchers aim to improve the early detection and intervention of this neurodegenerative disorder [23].

Mengting et al. in their study to classify EEG signals using a CNN architecture-based method, the proposed approach involves data preprocessing to enhance data quality by removing noise, artifacts, and other disturbances through resampling, bandpass filtering, and artifact removal. Subsequently, a five-layer CNN architecture is employed, comprising layers such as Pooling, Convolution, and Fully Connected layers, to extract and categorize features for classification. The HMT algorithm based on this CNN architecture was evaluated on a dataset containing four categories of EEG signals, demonstrating superior accuracy and stability compared to traditional methods. Specifically, the average accuracy rates are 81.5% for LDA, 86.2% for SVM, 89.6% for MLP, and 92.5% for HMT. In conclusion, the HMT algorithm offers promise for EEG signal classification, outperforming conventional approaches like LDA, SVM, and MLP in classification and recognition tasks [18].

Dipti et al. use EEG data preprocessing, involved using independent component analysis (ICA) to remove artifacts from the signals. Feature extraction was performed using wavelet transform and connectivity features, with the wavelet features including statistical variables such as max, min, STD, mean, and energy at each decomposition level. The classification was performed using a SVM classifier, which locates a hyperplane within the feature space by maximizing the distance between each class's nearest data point and the hyperplane. The proposed method, combining DWT and MaxLCoR features, showed a significant improvement in classification accuracy compared to using independent features, with an average accuracy of $40.64 \pm 2.45\%$ in a five-class classification setting [24].

Chen et al. in their paper, list some EEG signal feature extraction and classification models. The method involves the use of deep learning models such as EEGNet with Depthwise and separable convolutions obtaining an AUC in average of 90.54%, LADD-EEGNet with LSTM and domain discriminator with ACC of 94.69%, TSGL-EEGNet with TCSGL with ACC of 78.96%, S-EEGNet with separable convolutions and HHT transform with ACC of 77.90%; NeuroGrasp with two-state deep learning framework with ACC of 86%, and TinySleepNet with representation learning with ACC of 87.5%. For feature extraction and classification in EEG signals [25].

Current state of the art studies have significant limitations because the EEG signals used are uniform in size. In contrast, the BrainLat database presents a more significant challenge because it contains signals of different lengths. Classification models in the state of the art are designed to receive data of uniform dimensions, which poses the problem of compressing or transforming the signals to match in dimensions. Although techniques such as the use of wavelets, decomposition, spectrograms, or transforms have proven to be effective [10–13,17,18,22–25], they have yet to be widely applied to problems with signals of different lengths. This implies that compression or transformation methods must be applied to all signals, bringing them to the smallest or largest dimension. However, if the smallest or largest signal belongs to only one class, these techniques may bias the data by changing the spatial or frequency shape of the signals to this single-class signal.

The Fourier spectrogram offers a solution by transforming all the signals into a new vector space. Since they only go to 40 Hz, all the resulting images will have the same height. Although the width of the images may vary, they will maintain dimensions that are very

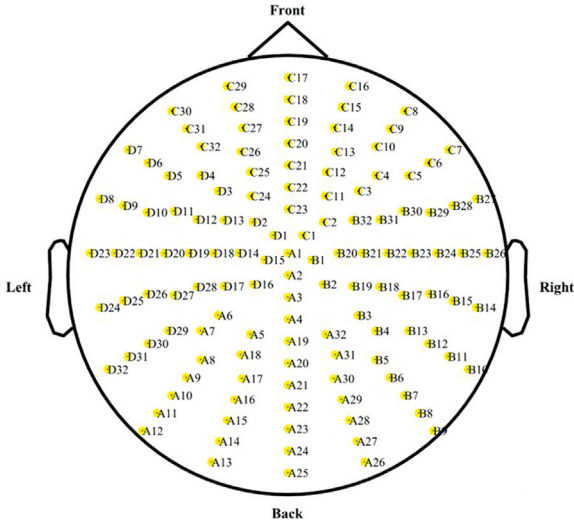


Fig. 1. Electrode layout of the Biosemi Active-two system [27].

close to the height according to the parameters used in the STFT calculation. Images whose width is unsuitable can be adjusted by cubic interpolation using OpenCV. In addition, state of the art models, such as CNNs, RNNs, LSTMs, and SVMs [10–13,17,18,22–25], only consider local features of the data. This limitation is solved by using visual transformers, which analyze local features and global relationships of the data. This allows much more information to be extracted, providing a much more robust classification and interpretation of the model than the techniques used in the state of the art.

3. Data description

3.1. Data base

This study used the BrainLat project database. BrainLat was launched in 2023 as a public-private partnership to test EEG, MRI, and fMRI, among other biomarkers and clinical tests. The dataset includes 530 patients with neurodegenerative diseases such as AD, bvFTD, MS, and PD, and 250 HCs. In addition, it includes demographic information about harmonized recruitment and assessment protocols. Participants were recruited from the Multi-Society Consortium clinical centers to Expand Dementia Research in Latin America (ReDLat), a regional effort to harmonize participant recruitment and neurocognitive assessment in multicenter studies [26].

EEG recordings and the classes AD, bvFTD, and HCs were used exclusively in this study. According to BrainLat, participants sat in a comfortable chair in a sound-attenuated, dimly lit, electromagnetically shielded room and were asked to remain still and awake. They recorded resting-state EEGs with eyes closed for ten minutes using a 128-channel Biosemi Active-two acquisition system, with reference electrodes placed on the mastoids. External electrodes were also placed at periocular locations to record eye blinks and eye movements. The analog filters used were set between 0.03 and 100 Hz. EEG was monitored online for drowsiness and myogenic and sweat artifacts. Data were re-referenced to the average of all channels, band-filtered between 0.5 and 40 Hz, downsampled to a sampling rate of 512 Hz, and ICA was used to correct for artifacts induced by flicker and eye movements. Defective channels were identified and replaced by weighted spherical interpolation [26].

The Biosemi Active-two acquisition system utilizes a 128-electrode configuration, as the Fig. 1 illustrates. For each patient, 128 EEG signals, see Fig. 2A, were recorded and subsequently processed in this study using the STFT. This method allows for generating frequency

spectrograms from each EEG signal, providing insights into the temporal occurrence of various brain waves. Consequently, 128 spectrograms were produced for each patient, which were then organized into layers, forming a $224 \times 224 \times 128$ matrix, as depicted in the Fig. 2.

3.2. Data-preprocessing

The EEG signals were extracted from the RAW format using Python with the MNE library, which facilitates data decompression, signal plotting, and retrieval of information such as signal magnitude, time, sampling frequency, and channel names, See Fig. 2A. Once decompressed, the STFT was computed using the SciPy library with signal parameters $stft(data, fs, nperseg = 4 * fs, noverlap = 3 * fs, nfft = 5.6 * fs)$ as is illustrated in Fig. 2B, the STFT is defined as

$$X_{STFT}(a, f) = \int_{-\infty}^{+\infty} x(t)g^*(t-a)e^{-j2\pi ft} dt \quad (1)$$

In the above definition, $g(t-a)$ is a shifted version of a time window $g(t)$ that extracts a portion of the signal $x(t)$. In other words, the window $g(t-a)$, having a limited duration, selects and extracts only a portion of the signal $x(t)$ to be analyzed by the Fourier Transform (FT). This time window is usually a real-time function; hence, $g^*(t-a) = g(t-a)$.

In simple terms, in the STFT, a time window selects a portion of $x(t)$, and then the regular FT is calculated for this selected part of the signal. By changing the shift parameter a , the FT of each part of the signal is obtained, and the temporal location of each portion is extracted at known time intervals determined by the shift factor a . In other words, the STFT analyzes both temporal and frequency information of each selected signal portion.

However, as the above definition reveals, the STFT has two parameters, f and a . This implies that the process requires more computation than the FT.

This configuration balances time and frequency resolution, crucial for preserving as much spectral information as possible. The frequency resolution adjustment resulted in the y -axis having a height of 224, corresponding to 40 Hz, which aligns with the passband filter limit used by BrainLat for EEG data. To standardize the dimensions, only the x -axis was resized using OpenCV with the Inter-Cubic interpolation method, a standard approach. This resizing was necessary as the signals varied in length, with an average difference of approximately 3 min. For each channel, an EEG spectrogram was generated and then grouped to form a matrix with a depth of 128 layers per patient, resulting in a dimension of $224 \times 224 \times 128$, see Fig. 2C.

The signals were not truncated to ensure that potentially valuable segments containing critical information related to AD were retained. Additionally, no compression techniques, such as Dynamic Time Warping, were applied to maintain the integrity of the original information. This approach ensures that all relevant information is preserved in the spectrograms, allowing the CVT model to extract and learn from the comprehensive data. Finally, the dataset was split into 80% for training and 20% for testing

4. Spectrocvnet-net

Our architecture consists of two modules or stages. The first stage is feature extraction from the input spectrogram, reducing the feature map's dimension by applying different blocks. The second is the classification stage, where a visual transformer is used, and finally, a fully connected layer for classification. The complete schematic of the SpectroCVT-Net architecture is shown in Fig. 3 A. The input image has dimensions of $224 \times 224 \times 128$. After passing through the feature extraction blocks, it becomes $224 \times 224 \times N$, where N is the number of filters. Upon entering the CVT block and before entering the transformer encoder, the tensor is transformed to $14 \times 14 \times 512$. Exiting the transformer with the vectorized information, it becomes a 1×128 tensor, which then enters the fully connected layers.

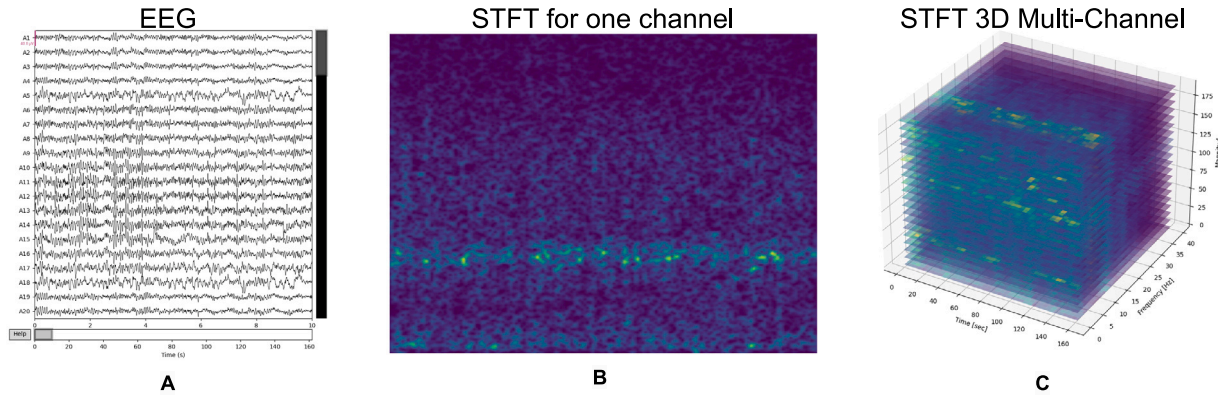


Fig. 2. Preprocessing of Signals. The figure consists of three parts: (A) the EEG signal for the first 20 electrodes, (B) the STFT of one channel from the EEG signal, and (C) the STFT of the 20 channels visualized in 3D, illustrating the stacking process to obtain a $224 \times 224 \times 128$ matrix.

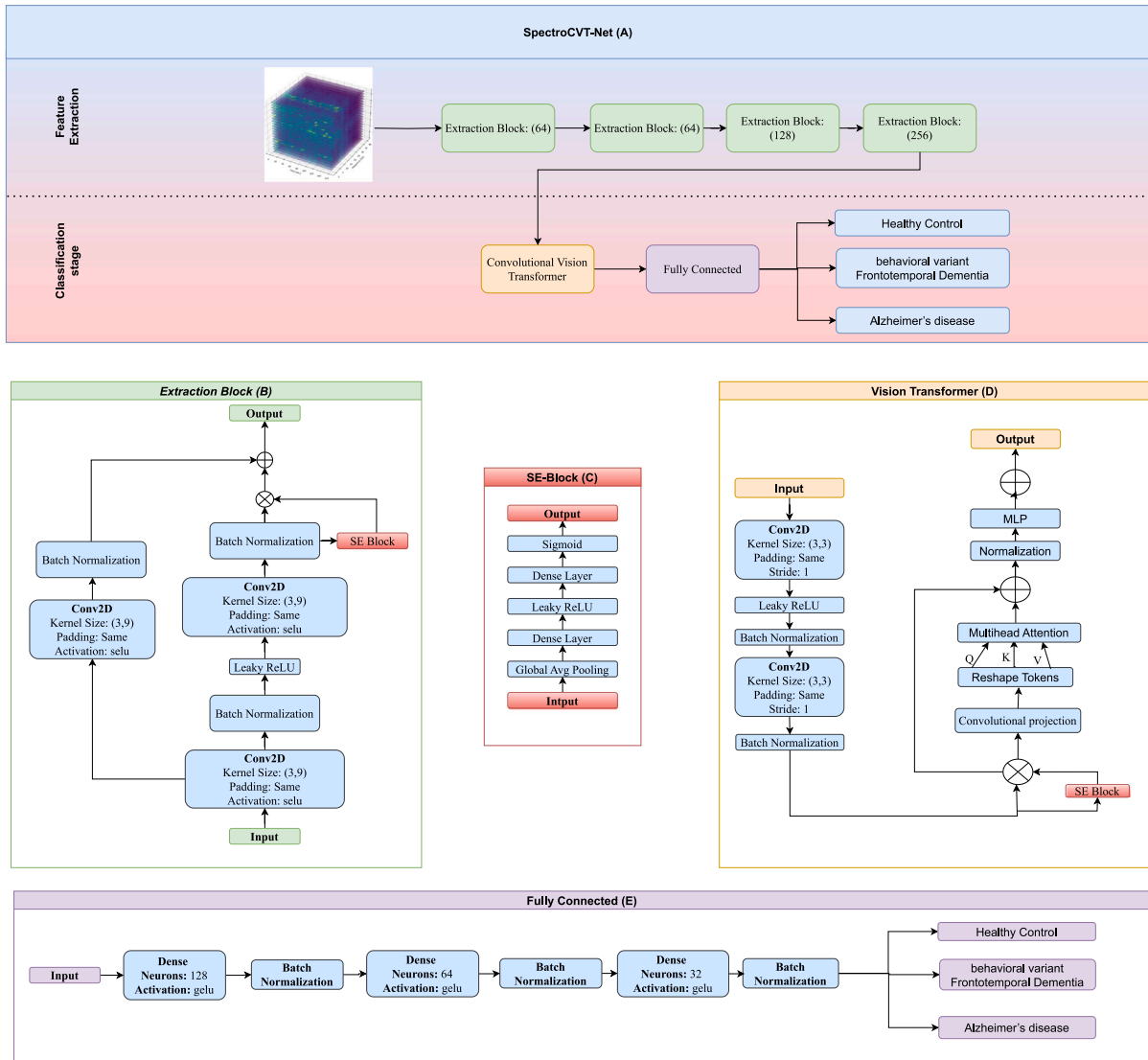


Fig. 3. (A) Complete architecture of the SpectroCVT-Net model. (B) Feature extraction block. (C) SE attention channel. (D) ViT architecture. (E) Fully connected and classification output.

4.1. Feature extraction

The feature extraction module comprises four extraction blocks, as illustrated in Fig. 3 B. Each block is divided into two sections and features a specific number of filters.

In the first section, the process begins with a 3×3 convolutional layer with a stride of 2×2 , followed by Batch Normalization and a Leaky ReLU activation, which allows non-linearities and permits negative values to pass with a slight slope. This is succeeded by another 3×3 convolutional layer with a stride of 1×1 , which maintains the spatial dimensions of the input, followed by another Batch Normalization. The resulting feature map is then multiplied by the SE-Block output, recalibrating the feature maps based on their relevance. The second section starts with a 3×3 convolutional layer with a stride of 2×2 , immediately followed by Batch Normalization. Finally, the outputs from both sections are summed, integrating features from both paths before proceeding to the next stage. This block includes Batch Normalization, defined by:

$$BN(x, \gamma, \beta) = \beta + \gamma \frac{x - E[X]}{\sqrt{Var[X] + \epsilon}} \quad (2)$$

where:

$E[X]$: expectation

$Var[X]$: variance

γ and β : scalars

The block also uses Leaky ReLU, defined as:

$$LeakyReLU(x) = \begin{cases} mx & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \quad (3)$$

where the negative slope m controls the linear function's decrease.

The extraction block's design combines convolutions, batch normalization, Leaky ReLU, and SE-Block to enable the network to learn robust and varied representations. The initial 3×3 convolutional layer with a stride of 2×2 reduces spatial dimensionality and computational load while capturing essential features. Batch Normalization improves stability and training speed, and Leaky ReLU allows gradient flow for negative values, mitigating the "vanishing gradient" problem. The second 3×3 convolutional layer with a stride of 1×1 allows granular feature extraction while maintaining spatial dimensions.

The SE-Block introduces attention, recalibrating feature maps according to their relevance, which enhances the model's focus on critical areas. The second section's 3×3 convolutional layer with a stride of 2×2 further reduces spatial dimensionality, and Batch Normalization ensures well-scaled and stabilized features. This section also functions as a skip connection, preventing gradient vanishing and integrating diverse features for a comprehensive representation.

4.1.1. SE-block

This architecture incorporates a Squeeze and Excitation layer (Se-Block), constructed using a two-dimensional Global Average Pooling, a dense layer with Leaky ReLU activation, and another dense layer with Sigmoid activation. The complete block is depicted in Fig. 3 C. It should be noted that all current classifying AD models based on DL predominantly use CNNs. These models extract features by incorporating spatial information and the information from each channel into the local receptive field. Usually, the data from all channels is combined during the convolution process; however, this approach is not optimal, as deep neural networks cannot learn discriminatively when each channel has the same weight. The Se-Block serves as a channel attention mechanism that allows the model to autonomously identify the channels and the importance of their features, i.e., their weights during the training process [28,29]. In essence, the primary goal of this block is to amplify the most significant channels and diminish the less important ones. The Se-Block is fundamental in our model since the input data to our architecture has 128 channels, and it is necessary to obtain the channel (electrode) that provides the most information to classify the disease. The squeeze and excitation operation yields weight parameters to be applied to the channels.

Squeeze operation. Encoding a spatial feature into a global function is referred to as the squeeze operation, achieved through Global Average Pooling, which decreases the dimensionality of the feature channels and streamlines subsequent computations, as illustrated in (4).

$$G = F_{sq}(M) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W M \quad (4)$$

where:

M : Feature map of size $H \times W \times C$

F_{sq} : Squeeze operation

G : Represent the output, the statistical information. It should also be made clear that $G \in R^C$

Excitation operation. The excitation operation is performed on the global feature. This operation begins with a dense layer with Leaky ReLU activation of dimension one, followed by a dense layer with Sigmoid activation with the respective block dimension. This process is described in (5).

$$M' = DL_{sm}(DL_{rl}(G)) \quad (5)$$

with:

M' : as the new feature where $M \in R^C$

DL_{rl} : as the Dense Layer with Leaky ReLU activation where $M \in R^{C/r}$

DL_{sm} : as the Dense Layer with Sigmoid activation

This block is utilized in other layers by multiplying the input with the weights obtained from the Se-Block.

4.2. Classification stage

4.2.1. Convolutional vision transformer

ViT networks have shown impressive results in various computer vision tasks in recent years. The convolutional Transformer retains the benefits of the traditional ViT while incorporating the advantages of convolutional networks, allowing it to model spatial relationships by capturing global dependencies. In this context, convolutional transformer networks can model local and global noise residuals in steganographic images. Fig. 3 D displays the internal structure of the CVT architecture. On the left side of the transformer architecture, certain operations consolidate feature maps while preserving the enhanced characteristics of previous blocks. This sequence includes a convolutional layer with dimensions 3×3 , a Batch Normalization layer, a Leaky ReLU activation function, another convolutional layer with dimensions 3×3 , and an additional Batch Normalization layer. The information then passes through an Se-Block on the right before entering the transformer architecture. Unlike the previous Se-Block, which uses Leaky ReLU, this Se-Block uses ReLU in the bottleneck. Within the transformer architecture, each block begins with the convolutional embedding of $2D$ tokens, enabling the feature maps to undergo reduced spatial sampling, thereby enriching the spatial representation of the tokens. The input tokens are reshaped into $2D$ before being processed by the convolutional transformer block, where convolutional projection operations are performed prior to self-attention to obtain the query, key, and value vectors $Q = K = V$, instead of the linear positional projection in ViT. The vectors are flattened to facilitate multi-head attention computation. The convolutional projection includes a multi-head attention block and an MLP with an added normalization layer before each [30,31].

Positional embedding. The convolution operations in token embedding and projection effectively capture spatial features from both global and local perspectives. The proposed network employs learnable $1D$ positional embeddings, which are added to the token embeddings in the convolutional Transformer (refer to Fig. 3).

Convolutional projection. The convolutional operation aims to model spatial and local steganographic noise at various semantic levels, from low-level edges to higher-order features, using a hierarchical, multi-stage approach similar to CNNs. At each stage, given an input x_{i-1} in the form of a 2D image or a 2D-reshaped token map from a previous stage with dimensions $x_{i-1} \in \mathbb{R}^{H_{i-1} \times W_{i-1} \times C_{i-1}}$, a 2D convolution operation $f(\cdot)$ is learned, mapping x_{i-1} to new tokens $f(x_{i-1})$ with channel size C_i . The kernel size is $s \times s$, and the stride is $s - o$, with p padding [19]. For the resulting token map $f(x_{i-1})$, the height and width are represented by $f(x_{i-1}) \in \mathbb{R}^{H_i \times W_i \times C_i}$, as shown in (6). $f(x_{i-1})$ is then flattened and normalized to a size of $H_i W_i \times C_i$ [19].

$$H_i = \left\lceil \frac{H_{i-1} + 2p - s}{s - o} + 1 \right\rceil, W_i = \left\lceil \frac{W_{i-1} + 2p - s}{s - o} + 1 \right\rceil. \quad (6)$$

Through the convolution projection, it can adjust the parameters of the convolution operation, allowing us to modify the feature dimension of tokens and the number of tokens at each stage. The proposed block transformer with convolutional projection extends the original block transformer by incorporating local spatial context modeling and improving efficiency by enabling undersampling of K and V matrices using depth-separable convolutional layers to implement multi-head self-attention. The projected tokens are ultimately compressed into a 1D format for subsequent processing. The projected token is formulated as follows,

$$x_i^{q/k/v} = \text{Flatten}(\text{Conv2d}(\text{Reshape2D}(x_i), s)) \quad (7)$$

where, $x_i^{q/k/v}$ represents the token input for matrices Q, K, and V at layer i , x_1 denotes the token before Convolutional Projection, and s indicates the convolution kernel size [19].

4.2.2. Fully connected layer

The architecture concludes with a fully connected layer consisting of Dense layers and Leaky ReLU. The dimensions of the Dense layers decrease from 128 to 3, with Softmax applied to make predictions and finalize the model. Refer to Fig. 3 E.

Softmax. An instance x , the softmax regression model first calculates a score $sk(x)$ for each class k and then estimates the probability of each class by applying the softmax function (also known as normalized exponential) to the scores. The equation for calculating $s_k(x)$ is similar to the equation for prediction in linear regression. Each class has its parameter vector $\theta^{(k)}$, typically stored as rows in a parameter matrix Θ . After calculating the score of each class, for instance, x , one can estimate the probability pk that the instance belongs to class k by passing the scores through the softmax function. This function calculates the exponential of each score and then normalizes them by dividing by the sum of all exponentials [32],

$$\hat{P}k = \delta(S(x)_k) = \frac{\exp S_k(x)}{\sum_{j=1}^K S_j(x)} \quad (8)$$

where:

K is the number of classes

$s(x)$ is a vector containing the scores of each class for the instance

$\sigma(s(x))_k$ is the estimated probability that instance x belongs to class k , given the scores of each class for that instance.

4.2.3. Hyperparameters

The hyperparameter optimization involved an exhaustive search within defined ranges to determine the best ones for the model. First, the hyperparameters corresponding to the feature extraction block were optimized, testing different activation methods such as SELU, RELU, and GELU, and finally selecting the Negative RELU. Following this, the hyperparameters of the transformer, including the number of attention heads and the number of layers, were optimized within a range and step size, as shown in Table 1.

Next, the model output was optimized, specifically the fully connected layers. Initially, the layers were arranged in descending powers

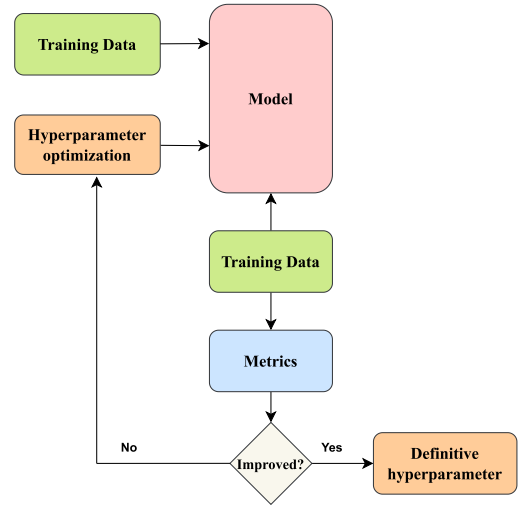


Fig. 4. The figure illustrates the hyperparameter optimization process in machine learning models. It shows the training cycle with data, metrics evaluation, and hyperparameter tuning until a definitive improvement is achieved.

of 2, starting from 1024 neurons down to 32 neurons, but eventually, only three layers were retained: 128, 64, and 32 neurons. Various activation methods were tested, ultimately selecting RELU. The optimization of the activation method involved testing simple Adam and Stochastic Gradient Descent, ultimately choosing adaptive Adam starting at 0.001, with decay steps of 100,000 and a decay rate of 0.96.

The optimization of these hyperparameters was conducted in an experimental and iterative manner, where each hyperparameter was adjusted, and the model was trained to observe the performance metrics, as illustrated in Fig. 4.

5. Methods

5.1. Transfer learning

5.1.1. VGG16

The VGG-16 model is a CNN architecture proposed by the Visual Geometry Group (VGG) at the University of Oxford. The architecture is defined as an input layer with $224 \times 224 \times 3$ dimensions. A convolutional layer with two consecutive convolutional layers with 64 filters each and a filter size of 3×3 . The same padding is applied to maintain spatial dimensions. A Max Pooling Layer with a pool size of 2×2 and a stride of 2. Two consecutive convolutional layers with 128 filters each and a filter size 3×3 . Max-pooling layer with a pool size of 2×2 and a stride of 2. Two consecutive convolutional layers with 256 filters each and a filter size 3×3 . Two sets of three successive convolutional layers with 512 filters each and a filter size 3×3 . Max-pooling layer with a pool size of 2×2 and a stride of 2. Two additional convolutional layers after the previous stack with a filter size of 3×3 . Flatten the output feature map $7 \times 7 \times 512$ into a vector of size 25088. Three fully connected layers with ReLU activation consist of a First layer with input size 25088 and output size 4096, a Second layer with input size 4096 and output size 4096, a Third layer with input size 4096 and output size 1000, corresponding to the 1000 classes in the ILSVRC challenge, a Softmax activation is applied to the output of the third fully connected layer for classification [33].

5.1.2. VGG19

VGG19 is a CNN renowned for its simplicity and effectiveness in image recognition tasks. With 19 layers, it follows a straightforward design principle of stacking multiple 3×3 convolutional layers, followed by max-pooling layers for down-sampling. This repetitive structure

Table 1

Hyperparameters used in the SpectroCVT-Net architecture, along with a description of each hyperparameter, the range, and the step at which they were iterated for optimization.

Hyperparameter	Description	Value	Range	Step
LAYER_NORM_EPS	This hyperparameter is used in layer normalization. EPS (epsilon) is a small value added to avoid division by zero during normalization.	1e-6	1e-5–1e-1,	1e-n, n=1,2,...,6
PROJECTION_DIM	This is the size of the projection dimension in the model. Specifically, it is the dimension to which the data is projected before entering the attention block or subsequent layers.	128	64–256	2 ⁿ
NUM_HEADS	This hyperparameter indicates the number of heads in the multi-head attention mechanism. Having multiple heads allows the model to focus on different parts of the input simultaneously.	2	2–16	4
NUM_LAYERS	This parameter defines the number of layers in the network, particularly in the transformer block or attention block.	2	2–16	4
MLP_UNITS	Defines the units in the multi-layer perceptron (MLP) network after the attention layer.	16	2–16	2
LEARNING_RATE	This is the hyperparameter that controls the learning rate of the optimizer, i.e., the magnitude of adjustments made to the model weights during training.	1e-3	1e-5–1e-1,	1e-n, n=1,2,...,6
WEIGHT_DECAY	This hyperparameter is used in the optimizer to prevent overfitting. It is a penalty applied to large weights during training, encouraging the model to maintain smaller weights.	1e-4	1e-5–1e-1,	1e-n, n=1,2,...,6
lr_schedule	The learning rate schedule is a technique that adjusts the learning rate during training instead of keeping it constant. In this case, an ExponentialDecay schedule is being used, which decreases the learning rate exponentially over time.	Lr_initial= 0.001	1–0.001	1e-n, n=1,2,3
Batch_Size	The number of samples processed before the model's parameters are updated. In other words, it is the number of training examples used to calculate the gradient of the model's parameters once before performing an optimization update.	128	64–256	2 ⁿ , n=6,7,8
Epochs	Refers to one complete cycle through the entire training dataset. This means that in one epoch, the model has seen each training sample once.	2 class= 150; 3 class= 500	100–500	50

results in a deep network capable of capturing intricate features at different levels of abstraction. VGG19's uniform architecture makes it easy to understand and implement, contributing to its popularity. In enhancing the perception capabilities of autonomous vehicles, VGG19's depth and feature-extraction capabilities play a crucial role in semantic segmentation, enabling the cars to accurately perceive and navigate complex real-world scenarios [33].

5.1.3. InceptionResNetV2 and inceptionV3

InceptionV1 used convolutions such as 5×5 , which caused the input dimensions to decrease by a large margin. This reduces accuracy because the neural network is susceptible to information loss if the input dimension decreases too drastically. In the InceptionResNetV2 architecture, the 5×5 convolution was replaced by two 3×3 convolutions. This also reduced computational time and increased computational speed because a 5×5 convolution is 2.78 times more expensive than a 3×3 convolution. Using two 3×3 layers instead of a 5×5 increases the architecture's performance. This architecture also converts $n \times n$ factorization into $1 \times n$ and $n \times 1$ factorization. As discussed above, a 3×3 convolution can be converted into a 1×3 followed by a 3×1 convolution, which is 33% cheaper in terms of computational complexity compared to a 3×3 [34,35].

To deal with the problem of the representational bottleneck, the feature banks of the module were expanded instead of making it deeper. This prevents the loss of information that occurs when making it deeper. InceptionV3 is similar to InceptionResNetV2 and contains all its features with the following changes/additions: use of the RMSprop

optimizer; batch normalization in the fully connected layer of the auxiliary classifier; use of 7×7 factorized convolution; label smoothing regularization, which is a method to regularize the classifier by estimating the effect of label dropout during training. It prevents the classifier from predicting a class too confidently. Adding label smoothing gives a 0.2% improvement in the error rate [34,35].

5.1.4. DenseNet201

DenseNet201 architecture is based on a series of dense blocks, each containing multiple convolutional layers. Each dense block takes the output of the previous block as input and the outputs of all the previous blocks. The core idea is that the feature maps produced by each layer can be concatenated to form the input to the next layer. This means that the output of each layer is the input of all the following layers. DenseNet201 also uses a transition layer between each dense block to reduce the spatial dimensionality and the number of feature maps. The transition layer consists of a batch normalization layer, a 1×1 convolutional layer, and a pooling layer. DenseNet201 has 201 layers, making it a larger and more complex model with almost twice the number of parameters compared to DenseNet121. Consequently, DenseNet201 requires more computation during the training and inference phases, making it more computationally expensive [36].

5.1.5. EfficientNetB7

The total of layers in EfficientNetB7 is 813; the layers that conform to the architecture can be made from 5 modules; the first module is composed of DepthWise Conv2D, Batch Normalization, and Activation; the second module of DepthWise Conv2D, Batch Normalization,

Activation, Zero Padding, Depthwise Conv2D, Batch Normalization and; the third module of Global Average Pooling, Recalling, and the layers of Conv2D; the fourth module of Multiply, Conv2D and Bath Normalization; and finally the five module of Multiply, Conv2D, Bath Normalization and Dropout. Module 1 is used as a starting point for the sub-blocks. Module 2 is a starting point for the first sub-block of all the seven main blocks except the first one. Module 3 is connected as a skip connection to all the sub-blocks. Module 4 is used to combine the skip connections in the first sub-blocks. Module 5 connects each sub-block to its previous sub-block in a skip connection, and they are combined using this module. These modules are further combined to form sub-blocks, which will be used in a certain way in the blocks. Sub-block 1 is used only as the first sub-block in the first block. Sub-block 2 is used as the first sub-block in all the other blocks. Sub-block 3 is used for any sub-block except the first one in all the blocks [37].

5.1.6. ConvNeXtXLarge

ConvNeXtXLarge is a convolutional neural network model designed to take advantage of recent advances in computer vision architectures and improve the performance of traditional ConvNets. Introduced in 2022, ConvNeXtXLarge is based on simple but effective principles, such as using deep, expandable convolutional blocks, increased utilization of normalization and activation layers, and integrating modern scaling techniques. This architecture demonstrates that ConvNets can meet and even exceed the performance of newer models, such as Transformers, in computer vision tasks while maintaining computational efficiency with careful tuning and design improvements. ConvNeXtXLarge has shown competitive results in standard benchmarks such as ImageNet, standing out for its ability to handle high-resolution images and adaptability to various vision applications [38].

5.2. Fully connected

The fully connected classification layer connected to the transfer learning networks is a 2-layer network: one with 256 units and one with 128 units. Both layers use batch normalization methods, ReLU activation, and an Adamax optimizer in the output layer with categorical loss. The experiments for the 3-class classification were conducted with a batch size of 128. For the 2-class classification, the experiments ran for 150 epochs with a batch size of 128.

5.3. Metrics

Tabares-Soto et al. [39]. highlight the significance of using metrics in model evaluation, emphasizing the importance of distinguishing between false positives (FP), false negatives (FN), true positives (TP), and true negatives (TN). The definitions and applications of the most critical metrics are detailed below.

5.3.1. Accuracy

Accuracy is a measure ranging from 0 to 1, representing the percentage of correct predictions made by the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

5.3.2. Precision

Precision indicates the proportion of true positives over total positive predictions, helping to identify class distribution accurately.

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

5.3.3. Sensitivity (recall)

Also known as sensitivity, recall shows the classifier's ability to correctly identify positive instances in the data set.

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

5.3.4. F1 score

F1 Score is a metric that balances precision and recall, useful for addressing class imbalance.

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (12)$$

5.3.5. Support

Support indicates the number of instances in each test class, which provides an idea of the classes' size and representatives in the data set.

5.3.6. Confusion matrix

The confusion matrix is a tool that shows the relationship between the actual classes and those predicted by the model. The rows represent the predicted classes, while the columns represent the actual classes.

5.3.7. Cross-validation

Cross-validation, a technique used to evaluate model performance, is a practical and widely-used method. It divides the data into several subsets called 'folds' (k) of similar sizes, allowing us to obtain multiple training and test sets. This helps us get a more comprehensive understanding of our model's performance and its generalizability.

$$Cross-Validation = \frac{1}{k} \sum_{i=1}^k Performance_i \quad (13)$$

These metrics provide a holistic understanding of the model's performance, allowing us to evaluate its accuracy, generalizability, and handling of class imbalances.

5.4. Hardware and resources

Google Colab was used for all experiments. The experiments utilized two different GPUs: an NVIDIA T4 GPU with 15 GB of memory, CUDA Version 10.1, and 12 GB of RAM for general experiments, and an NVIDIA A100 GPU with 40 GB of memory for models EfficientNetB7, DenseNet201, and ConvNeXtXLarge, providing superior performance with enhanced capabilities. TensorFlow 2.15.0 and Python 3.10.12 were used for all tests.

6. Results

6.1. Parameters

The BrainLat database is very recent and therefore lacks other results, especially with the methodology used. For this reason, transfer learning networks were employed to compare the SpectroCVT-Net model and the obtained results. Firstly, we observe the number of parameters in Fig. 5, where the number of parameters for each model is compared. It is evident that the proposed model has the fewest parameters, with around 12 million, whereas ConvNeXtXLarge has the most parameters, with around 200 million. This demonstrates that larger networks are not always more powerful, especially due to the attention mechanisms, which result in less analysis but greater network efficiency.

6.2. Binary classification

The model comparison was conducted by class. When evaluating the model between the classes HC vs. bvFTD, the results are shown in Table 2. This table includes the model used, the experiment referred to as "folds" (where the partition seed is changed to achieve cross-validation), and the average of the folds. The proposed model stands out, achieving perfect classification in each fold, outperforming other transfer learning models.

On the other hand, when classifying against Alzheimer's, the model tends to make more errors. This is evident in Tables 3 and 4, where we classify HC vs. AD and bvFTD vs. AD. Although the proposed model

Table 2

Results for the HC vs bvFTD class. The table shows the models used for comparison, the metrics, the results for each of the folds, and finally the average of each metric across the 5 folds.

Model	Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
VGG16	Accuracy	36.36%	63.63%	63.63%	36.36%	36.36%	41.18% $\pm 0.10\%$
	Precision	18.18%	31.18%	31.18%	18.18%	18.18%	20.90% $\pm 5.45\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
VGG19	Accuracy	63.63%	63.63%	63.63%	63.63%	36.36%	52.72% $\pm 13.36\%$
	Precision	31.18%	31.18%	31.18%	18.18%	18.18%	26.36% $\pm 6.68\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
InceptionResNetV2	Accuracy	63.63%	63.63%	63.63%	36.36%	36.36%	57.72% $\pm 8.16\%$
	Precision	31.18%	31.18%	31.18%	18.18%	18.18%	26.36% $\pm 6.68\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
DenseNet201	Accuracy	63.63%	36.36%	63.63%	36.36%	36.36%	47.27% $\pm 13.36\%$
	Precision	31.18%	18.18%	31.18%	18.18%	18.18%	23.63% $\pm 6.68\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
EfficientNetB7	Accuracy	36.36%	63.63%	63.63%	63.63%	63.63%	58.18% $\pm 10.9\%$
	Precision	18.18%	31.18%	31.18%	31.18%	31.18%	29.09% $\pm 5.45\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
ConvNeXtLarge	Accuracy	63.63%	63.63%	63.63%	63.63%	63.63%	63.63% $\pm 0.00\%$
	Precision	31.18%	31.18%	31.18%	31.18%	31.18%	31.18% $\pm 0.00\%$
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm 0.00\%$
SpectroCVT-Net	Accuracy	100%	100%	100%	100%	100%	100% $\pm 0.00\%$
	Precision	100%	100%	100%	100%	100%	100% $\pm 0.00\%$
	Recall	100%	100%	100%	100%	100%	100% $\pm 0.00\%$

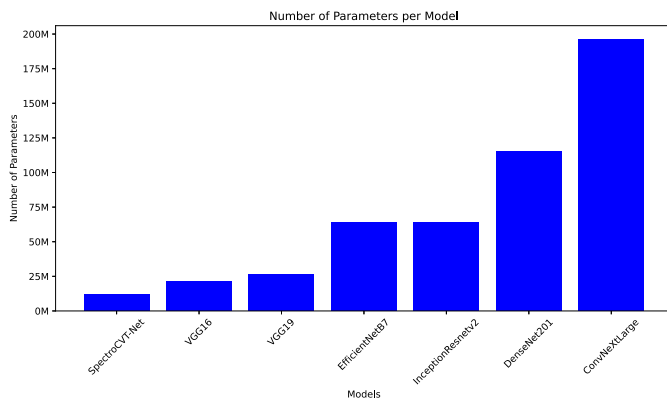


Fig. 5. Number of parameters per model. This bar chart compares the number of parameters across various neural network models used in the study. The models included are SpectroCVT-Net, VGG16, VGG19, EfficientNetB7, InceptionResNetV2, DenseNet201, and ConvNeXtLarge. The SpectroCVT-Net model has the fewest parameters, around 12 million, while ConvNeXtLarge has the most, approximately 200 million. This comparison demonstrates that larger networks are not necessarily more powerful, highlighting the efficiency of the SpectroCVT-Net model due to its attention mechanisms, which result in less analysis but greater network efficiency.

performs better, differentiating between dementia and Alzheimer's remains a challenge, as shown in the results obtained.

Finally, the binary classification metrics are shown in Fig. 6, which includes the classification report and the confusion matrix for each of the two-class divisions evaluated.

6.3. Multiclass

When evaluating the three classes HD, bvFTD, and AD, the proposed model performs better than the transfer learning models, as shown in Table 5. Similar to the binary partition, this table includes the models used, the folds, and the average of the folds with their respective standard deviation.

However, it is important to highlight that, as shown in Fig. 7, the confusion matrix indicates that the classification of Alzheimer's is the most challenging. Despite this, we believe that the proposed

model, with this methodology, is useful and can make a significant step forward in the classification of AD. All these experiments were performed under equal conditions: for three classes, a batch size of 128 and 500 epochs was used, while for the two-class tests, a batch size of 128 and 150 epochs was used.

6.4. Interpretation

This research evaluated the effectiveness of SpectroCVT-Net against other transfer learning networks. This was the first study to use the BrainLat database [26]. The experiments were performed by dividing the BrainLat database into training, validation, and test sets, which included 68 training spectrograms and 18 validation and test spectrograms, all experiments with K-Fold validation.

The network's effectiveness is demonstrated to be superior to transfer learning methods. However, it is crucial to provide interpretability to the network. Therefore, a Grad-CAM as shown in Fig. 8.

However, even though the Grad-CAM for each class is shown in a three-dimensional format, it is not clear which parts of our spectrogram are the most important for the model. Therefore, a heatmap of 128 channels for each Grad-CAM was created for each patient. This was achieved by averaging the images of all patients by class to identify the most important points of attention for the model. This can be seen in Fig. 9.

The results showed that:

- For AD, the layer with the highest pixel intensity is number 19, corresponding to channel A19, See Fig. 9A and 11A.
- For bvFTD, the layer with the highest pixel intensity is number 1, corresponding to channel A1 as illustrated in Fig. 9C and 11C.
- For HC, the layer with the highest pixel intensity is number 117, corresponding to channel D21, See Fig. 9B and 11B.

All layers were summed and divided by the total number of layers to normalize and understand which frequencies the network is most focused on.

This produced an average of all Grad-CAMs by class, allowing us to observe that the model puts its attention on:

- Class AD: Frequencies between 30 and approximately 38 Hz (Gamma waves) and between 0 and approximately 12 Hz (Delta,

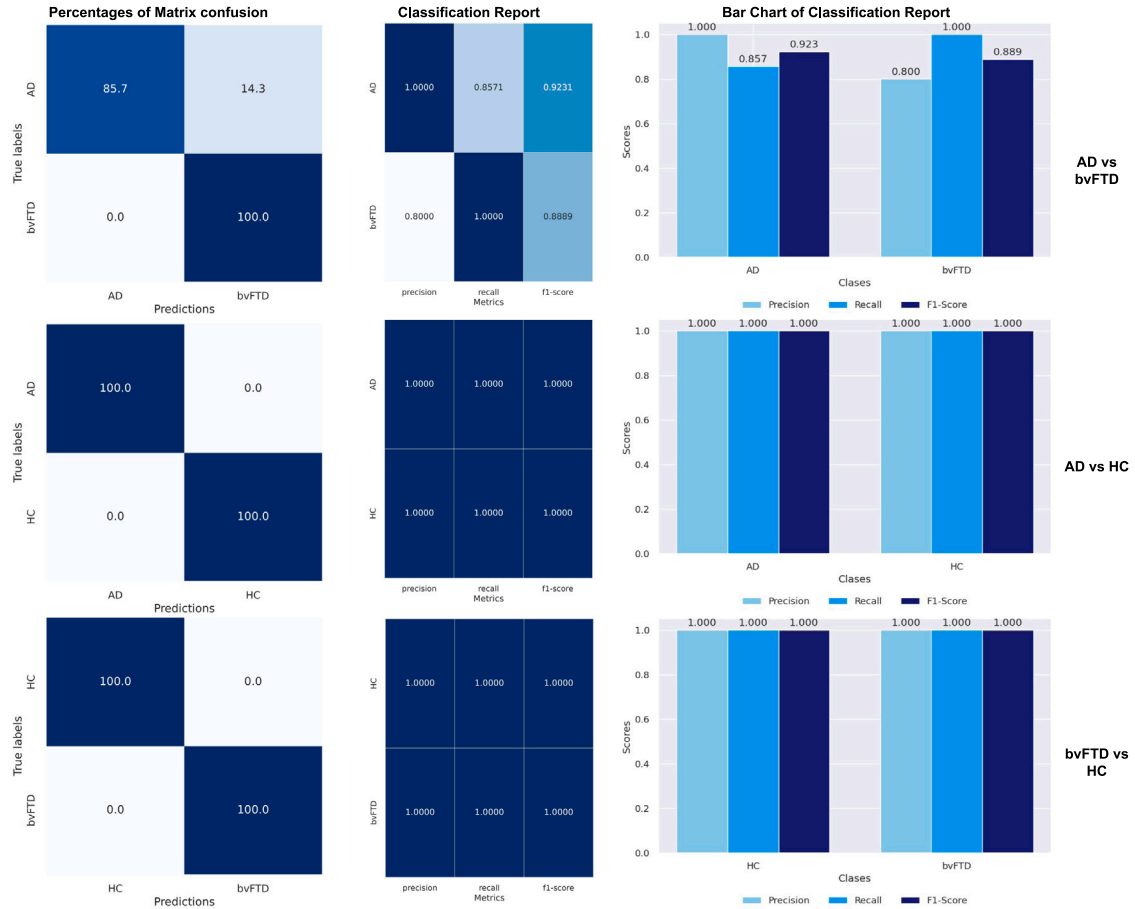


Fig. 6. Comparative table of the metrics of the best model for classification of two classes. The figure shows in the first column, the classification report; the second, the percentages of the confusion matrix; and the third, the Bar chart of the classification report.

Table 3

Results for the HC Vs AD class. The table shows the models used for comparison, the metrics, the results for each of the folds, and finally the average of each metric across the 5 folds.

Model	Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
VGG16	Accuracy	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
	Precision	25.00%	25.00%	25.00%	25.00%	25.00%	25.00% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
VGG19	Accuracy	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
	Precision	25.00%	25.00%	25.00%	25.00%	25.00%	25.00% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
InceptionResNetV2	Accuracy	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
	Precision	25.00%	25.00%	25.00%	25.00%	25.00%	25.00% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
DenseNet201	Accuracy	36.36%	36.36%	63.63%	36.36%	36.36%	41.81% $\pm$ 10.90%
	Precision	18.18%	31.18%	18.18%	18.18%	18.18%	21.59% $\pm$ 5.90%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
EfficientNetB7	Accuracy	63.63%	63.63%	63.63%	63.63%	63.63%	63.63% $\pm$ 0.00%
	Precision	31.18%	31.18%	31.18%	31.18%	31.18%	31.18% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
ConvNeXtXLarge	Accuracy	63.63%	63.63%	63.63%	63.63%	63.63%	63.63% $\pm$ 0.00%
	Precision	31.18%	31.18%	31.18%	31.18%	31.18%	31.18% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
SpectroCVT-Net	Accuracy	100%	100%	100%	100%	92.59%	98.51% $\pm$ 2.96%
	Precision	100%	100%	100%	100%	88.88%	97.77% $\pm$ 4.44%
	Recall	100%	100%	100%	100%	88.88%	97.77% $\pm$ 4.44%

Theta, Alpha waves) [16], with a greater focus on Delta waves as illustrated in Fig. 10A.

• Class bvFTD: Frequencies between 0 and approximately 15 Hz (Delta, Theta, Alpha, and lower Beta waves) [16], See Fig. 10C.

Table 4
Results for the bvFTD Vs AD class. The table shows the models used for comparison, the metrics, the results for each of the folds, and finally the average of each metric across the 5 folds.

Model	Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
VGG16	Accuracy	36.00%	64.00%	64.00%	64.00%	64.00%	58.00% $\pm$ 11.00%
	Precision	18.00%	32.00%	32.00%	32.00%	32.00%	29.00% $\pm$ 5.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
VGG19	Accuracy	36.00%	64.00%	64.00%	64.00%	64.00%	53.00% $\pm$ 11.00%
	Precision	18.00%	32.00%	32.00%	32.00%	32.00%	29.00% $\pm$ 5.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
InceptionResNetV2	Accuracy	36.00%	64.00%	36.00%	64.00%	64.00%	53.00% $\pm$ 13.00%
	Precision	18.00%	32.00%	18.00%	32.00%	32.00%	26.00% $\pm$ 7.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
DenseNet201	Accuracy	36.36%	36.36%	36.36%	63.63%	36.36%	41.81% $\pm$ 10.9%
	Precision	18.18%	18.18%	18.18%	31.18%	18.18%	20.90% $\pm$ 5.45%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
EfficientNetB7	Accuracy	63.63%	63.63%	63.63%	63.63%	63.63%	63.63% $\pm$ 0.00%
	Precision	31.18%	31.18%	31.18%	31.18%	31.18%	31.18% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
ConvNeXtXLarge	Accuracy	63.63%	63.63%	63.63%	63.63%	63.63%	63.63% $\pm$ 0.00%
	Precision	31.18%	31.18%	31.18%	31.18%	31.18%	31.18% $\pm$ 0.00%
	Recall	50.00%	50.00%	50.00%	50.00%	50.00%	50.00% $\pm$ 0.00%
SpectroCVT-Net	Accuracy	82.74%	82.74%	91.59%	91.59%	82.74%	85.59% $\pm$ 4.44%
	Precision	81.81%	81.81%	90.90%	90.90%	90.90%	85.45% $\pm$ 4.44%
	Recall	81.81%	81.81%	90.90%	90.90%	90.90%	85.45% $\pm$ 4.44%

Table 5
Results for the HC Vs bvFTD Vs AD class. The table shows the models used for comparison, the metrics, the results for each of the folds, and finally the average of each metric across the 5 folds.

Model	Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
VGG16	Accuracy	38.88%	38.88%	38.88%	22.22%	38.88%	35.55% $\pm$ 6.66%
	Precision	12.96%	12.96%	12.96%	7.40%	12.96%	11.85% $\pm$ 2.22%
	Recall	33.33%	33.33%	33.33%	33.33%	58.83%	33.33% $\pm$ 0.00%
VGG19	Accuracy	38.88%	38.88%	38.88%	22.22%	38.88%	35.55% $\pm$ 6.66%
	Precision	12.96%	12.96%	12.96%	7.40%	12.96%	11.85% $\pm$ 2.22%
	Recall	33.33%	33.33%	33.33%	33.33%	58.83%	38.33% $\pm$ 0.005%
InceptionResNetV2	Accuracy	38.88%	22.22%	38.88%	22.22%	22.22%	28.84% $\pm$ 8.16
	Precision	12.96%	7.40%	12.96%	7.40%	7.40%	9.62% $\pm$ 2.72%
	Recall	33.33%	33.33%	33.33%	33.33%	58.83%	38.33% $\pm$ 0.005%
DenseNet201	Accuracy	38.88%	22.22%	22.22%	28.88%	22.22%	28.88% $\pm$ 8.16%
	Precision	12.96%	7.40%	7.40%	12.96%	7.40%	9.62% $\pm$ 2.72%
	Recall	33.33%	33.33%	33.33%	33.33%	33.33%	33.33% $\pm$ 0.00%
EfficientNetB7	Accuracy	38.88%	38.88%	38.88%	38.88%	38.88%	38.88% $\pm$ 0.00%
	Precision	12.96%	12.96%	12.96%	12.96%	12.96%	12.96% $\pm$ 0.00%
	Recall	33.33%	33.33%	33.33%	33.33%	33.33%	33.33% $\pm$ 0.00%
ConvNeXtXLarge	Accuracy	38.88%	38.88%	38.88%	38.88%	38.88%	38.88% $\pm$ 0.00%
	Precision	12.96%	12.96%	12.96%	12.96%	12.96%	12.96% $\pm$ 0.00%
	Recall	33.33%	33.33%	33.33%	33.33%	33.33%	33.33% $\pm$ 0.00%
SpectroCVT-Net	Accuracy	92.59%	92.59%	92.59%	88.88%	96.29%	92.59% $\pm$ 2.34%
	Precision	88.88%	88.88%	88.88%	83.33%	94.44%	88.88% $\pm$ 3.35%
	Recall	88.88%	88.88%	88.88%	83.33%	94.44%	88.88% $\pm$ 3.35%

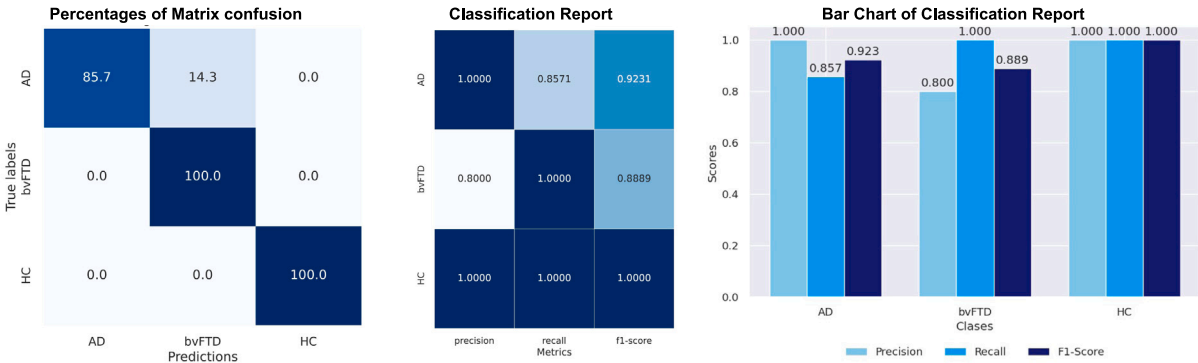


Fig. 7. Comparative table of the metrics of the best model for classification of the classes. In this figure the first column is the classification report, the second is the percentages of the confusion matrix, and the third is the Bar chart of the classification report.

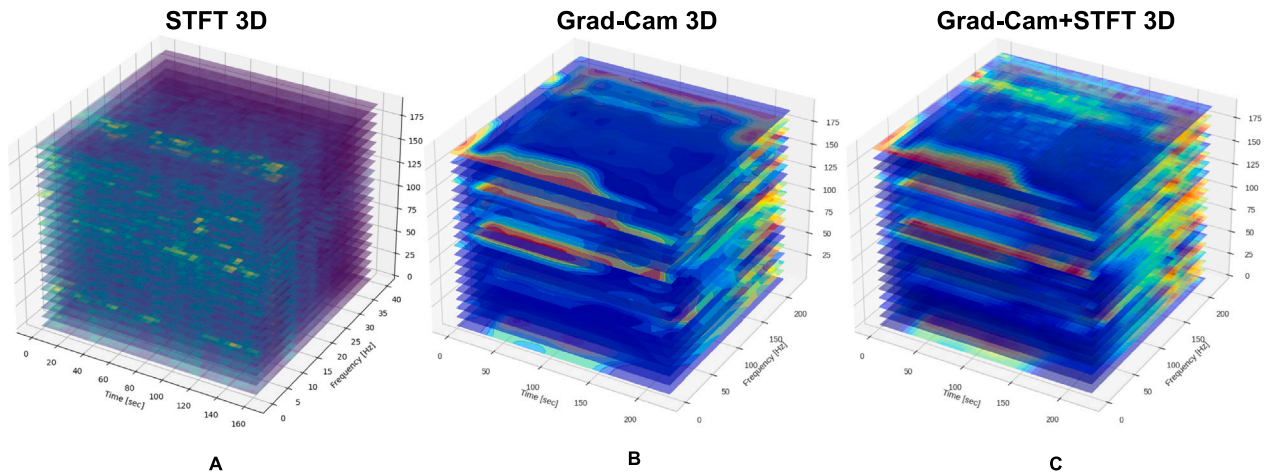


Fig. 8. 3D visualization of transformations and analysis of EEG signals using STFT and Grad-CAM. (A) Three-dimensional representation of the Short-Time Fourier Transform (STFT) applied to EEG signals. This visualization shows how frequencies are distributed over time for a specific EEG channel. (B) Three-dimensional visualization of the Grad-CAM technique applied to EEG signals. Grad-CAM highlights the regions and features most relevant to the model for classification, emphasizing the areas with the highest weight in the decision-making process. (C) Combination of STFT and Grad-CAM in a three-dimensional visualization. This combination provides an integral view of how frequencies are distributed over time and identifies the most important features according to the model, facilitating a deeper and more comprehensible interpretation of the data.

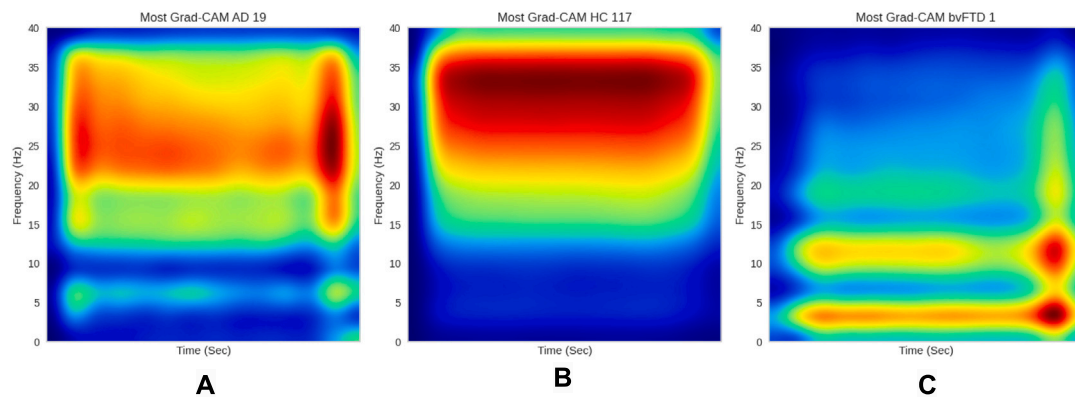


Fig. 9. Grad-CAM Focus Analysis. (A) Layer 19 of Grad-CAM for AD, where the model focused the most. (B) Layer 117 of Grad-CAM for HC, where the model focused the most. (C) Layer 19 of Grad-CAM for bvFTD, where the model focused the most.

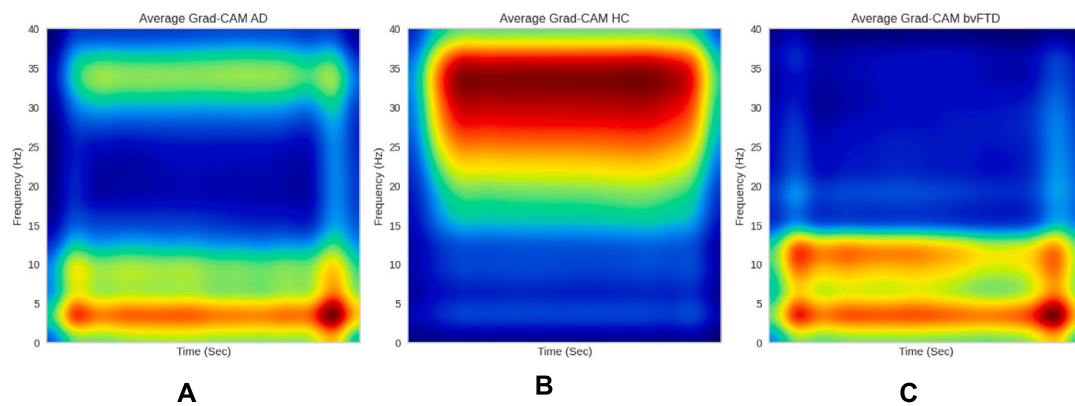


Fig. 10. Average Grad-CAM Focus. (A) The average Grad-CAM for the entire AD class highlights where the model focused the most. (B) The average Grad-CAM for the entire HC class shows the model's main focus areas. (C) The average Grad-CAM for the entire bvFTD class indicates the regions of greatest focus for the model.

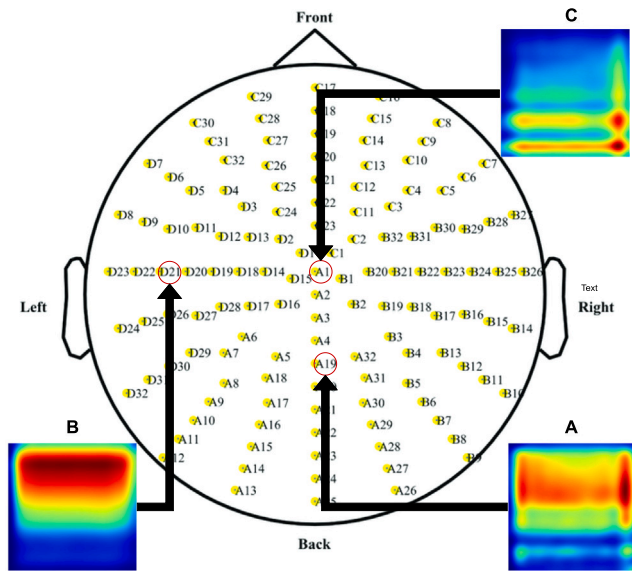


Fig. 11. Electrode Focus Points. (A) The location of electrode A19 corresponds to layer 19, where the model is focused the most for the AD class. (B) The location of electrode D21 corresponds to layer 117, where the model is focused the most for the HC class. (C) Location of electrode A1, corresponding to layer 1, where the model focused the most for the bvFTD class.

Source: taken and adapted from [27].

- Class HC: Frequencies between 15 and 40 Hz (Beta and Gamma waves) [16] as illustrated in Fig. 10B.

This interpretability methodology reveals the frequency ranges that provide the most information per class, allowing the model to make classification decisions. In addition, it provides researchers with an understanding of which electrodes provide the most information and the brain regions to which these electrodes correspond.

7. Discussion

Alzheimer's disease, as the most common form of dementia, affects approximately 55 million people worldwide. Diagnosing it presents a significant challenge, particularly in developing countries where cutting-edge equipment is either unavailable or difficult to access for many populations. This is why research into this pathology focuses on finding new mechanisms to assist medical personnel in reducing costs and time, thereby improving the patient's quality of life. Despite the relatively low cost, the choice of EEG spectrograms in this research is based on their ability to visualize properties and characteristics of EEG signals that are not evident in their temporal form. Spectrograms allow for a combined representation of temporal and frequency information, facilitating the identification of relevant patterns for diagnosing neurodegenerative diseases.

When comparing the proposed SpectroCVT-Net architecture Fig. 3 with various pre-trained models such as VGG16, VGG19, Inception-ResNetV2, DenseNet201, EfficientNetB7, and ConvNeXtXLarge, despite their effectiveness in other computer vision tasks and having a higher number of parameters Fig. 5, they were unable to match the performance of SpectroCVT-Net in classifying EEG spectrograms. This is attributed to the CVT's ability to capture global dependencies within the data, which is crucial for interpreting complex signals like EEG. A key aspect of SpectroCVT-Net's success is its combination of local convolutions with global attention mechanisms, allowing for a richer and more detailed representation of the relevant features for classification. Additionally, the incorporation of SE blocks enhances attention to the most informative channels, increasing the model's accuracy.

It is also important to explore new databases. BrainLat is a good starting point for testing different information sources since it is very common to use the ADNI or OASIS databases when discussing dementia in DL. Although they have sufficient information, different data sources from various parts of the world should be tested to avoid biases.

Considering the above and the functioning of SpectroCVT-Net, it is important to explain why the architecture works well. Therefore, interpretability in the 6.4 section is fundamental to this study as it analyzes the behavior of the different GradCAM obtained for each class Fig. 8 and their modifications when using GradCAM and the Fourier transform Fig. 9.

However, besides giving interpretability to the algorithm, it is also important to provide transparency. Fig. 12 shows the input–output flow of each data point. Here, we see how the model receives a complete spectrogram with dimensions $224 \times 224 \times 128$ and passes through various feature extraction blocks, gradually reducing in size to $112 \times 112 \times 64$, then to $56 \times 56 \times 64$, then $28 \times 28 \times 128$, and finally to an image of $14 \times 14 \times 256$ that enters our CVT model. In the CVT model, a convolutional projection begins, followed by its respective tokenization and attention map, and finally, it enters the fully connected layer to give an output.

It is important to discuss the clinical implications of these results. SpectroCVT-Net's ability to classify EEG signals with high accuracy has significant clinical implications. This approach could facilitate the early diagnosis of AD and bvFTD, enabling more timely and personalized interventions. Furthermore, the portability and low cost of EEG devices make this methodology accessible in resource-limited clinical settings, expanding its potential impact on global healthcare. Additionally, its interpretability and the ability to show which electrodes provide the most information Fig. 11 open the possibility for clinical research in this field and observing the results of these electrodes to provide an accurate diagnosis.

Despite the model's success, several challenges and limitations must be considered. Classifying Alzheimer's remains difficult, which could be due to the heterogeneous nature of the disease and the overlap of characteristics with other neurodegenerative conditions. Moreover, although the results obtained are promising, it is necessary to validate the model on larger and more diverse datasets to ensure its generalization and robustness. Another aspect to consider is the variability in the length of EEG signals. Although using spectrograms helps standardize image dimensions, adjusting the resolution through interpolation can introduce biases.

Finally, any research should include repositories to facilitate reproducibility, enhance the research's credibility, and advance the field.

8. Conclusion

This paper presents a novel architecture based on convolutional vision transformers and channel attention mechanisms for classifying Alzheimer's disease using EEG spectrograms. The SpectroCVT-Net architecture combines convolution and attention mechanisms to capture local and global dependencies within spectrograms, thereby enhancing classification accuracy. The results demonstrated an accuracy of $92.59\% \pm 2.3\%$ in classifying Alzheimer's disease, healthy controls, and behavioral variant frontotemporal dementia (bvFTD). The model evaluation also showed an accuracy of $98.51\% \pm 2.9\%$ for classifying Alzheimer's disease vs. healthy controls, $100\% \pm 0\%$ for bvFTD vs. healthy controls, and $85\% \pm 4.4\%$ for Alzheimer's disease vs. bvFTD. These findings underscore the effectiveness of SpectroCVT-Net in EEG spectrogram classification and its potential utility in the clinical diagnosis of neurological conditions such as Alzheimer's disease and frontotemporal dementia. Additionally, Grad-CAM analysis was used to provide greater interpretability, identifying critical layers, frequencies, and electrodes influencing the classification. Overall, this work contributes a new architecture specifically designed for EEG spectrogram classification without relying on transfer learning, offering a deeper understanding of brain waves associated with these diseases, which could aid in the clinical diagnosis and follow-up of these conditions.

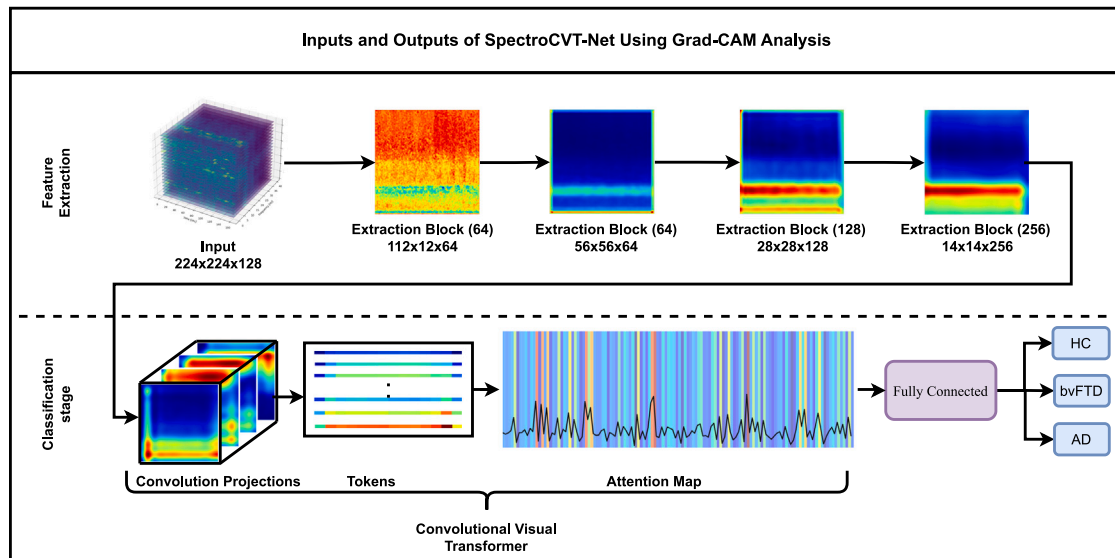


Fig. 12. Architecture of SpectroCVT-Net, featuring a series of extraction blocks with different dimensions, a convolutional visual transformer, and an attention map. It also includes Grad-CAM analysis outputs for healthy controls (HC), behavioral variant frontotemporal dementia (bvFTD), and Alzheimer's disease (AD).

8.1. Future works

This study has demonstrated the efficacy of the SpectroCVT-Net model in classifying Alzheimer's disease (AD), healthy controls (HC), and behavioral variant frontotemporal dementia (bvFTD) using EEG signal spectrograms. Despite the promising results, several areas require further investigation and development. Highlighting the importance of interdisciplinary collaboration among neurologists, engineers, and data scientists will be crucial to bring these innovations from the laboratory to the clinic. However, before transitioning to a real clinical setting, the following future works are proposed:

1. **Implementing Kolmogorov–Arnold Systems (KANs):** A significant future direction is the implementation of Kolmogorov–Arnold systems (KANs). These systems can help decompose complex functions into a sum of simpler functions, which could enhance the model's ability to capture non-linear relationships in EEG data, providing a richer and more accurate representation of brain signals.
2. **Implementing the Wavelet Transform:** Another area of future research is the implementation of the wavelet transform instead of STFT for generating spectrograms. The wavelet transform can offer better temporal and frequency resolution, potentially improving the model's accuracy in capturing finer and localized features of EEG signals.
3. **Enhancing Attention Mechanisms by Creating Custom Attention Modules:** Although the current SpectroCVT-Net architecture uses attention blocks, there is potential to improve these mechanisms by developing custom attention modules. These new attention modules could be specifically optimized for the nature of EEG signals, more effectively capturing critical spatial and temporal relationships in the data.
4. **Bimodal Analysis Image and Wavelet:** Transform The integration of bimodal data is a promising area for future research. In particular, combining brain images, such as MRI, with spectrograms generated by the wavelet transform can provide a more comprehensive view of brain activity and underlying pathologies. This combination could significantly enhance diagnostic accuracy and offer a more holistic approach to clinical assessment.

5. **Bimodal Analysis Wavelet and Fourier:** Besides combining different data modalities, it is also valuable to explore the integration of different signal analysis techniques. Combining spectrograms generated by the wavelet transform with those obtained through the Fourier Transform can capture a wider range of EEG signal characteristics. This bimodal approach can help identify patterns and features that a single technique might overlook, thereby improving the model's ability to classify different neurological states.

CRedit authorship contribution statement

Mario Alejandro Bravo-Ortiz: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ernesto Guevara-Navarro:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sergio Alejandro Holguín-García:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mariana Rivera-García:** Writing – original draft, Methodology, Investigation, Data curation. **Oscar Cardona-Morales:** Writing – review & editing, Writing – original draft, Supervision. **Gonzalo A. Ruz:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources. **Reinel Tabares-Soto:** Writing – review & editing, Writing – original draft, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and materials

The data supporting the findings of this study are available in BrainLat (Prado et al., 2023).

Repository: <https://github.com/SpectroCVT-Net>.

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